UNIT: The numerical value of any physical quantity is expressed as multiples of some standardized physical quantity. For instance height of a person could be expressed in multiples of inches, millimeters or feet etc. Such a standardized physical quantity is called a unit of measurement. For instance Km, Feet, millimeter and inches are all units of lengths and any length can be expressed as their multiple. Note that the physical quantity remains the same irrespective of how it is expressed in various units. For instance Height of a person does not change just because it is expressed in various units.

RELATION BETWEEN NUMERICAL VALUE AND UNIT: If A is the quantity of some physical variable and α_1 is the unit used to measure it then, the numerical value of measurement is given as $a_1 = A/\alpha_1$.

EFFECT OF UNIT CONVERSION ON NUMERICAL VALUES:

Let a physical quantity A be expressed in units α_1 and α_2 with corresponding numerical values a_1 and a_2 then by definition of units:

$$a_1 = \frac{A}{\alpha_1} \quad (1) \quad \text{and} \quad a_2 = \frac{A}{\alpha_2} \quad (2)$$

From (1) & (2):

$$\boxed{a_1\alpha_1 = a_2\alpha_2 = A}$$

Thus numerical value of a physical quantity is inversely proportional to the size of the unit used to measure it.

UNITS OF A QUOTIENT OR A PRODUCT: If A is measured in units α_1 and B is measured in units β_1 then A/B can be measured in units of α_1/β_1, similarly A×B can be measured in units of $\alpha_1 \times \beta_1$ for example if length is measured in meters and time in seconds then velocity can be measured

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

in units of m/s. (The same idea can be extended to products and quotients of powers of various physical quantities).

Numerical Value Of A Quotient Or A Product:

Let a physical quantity be expressed as $P = A/B$, then if P is expressed in units of α_1/β_1 and A is expressed in units of α_1 and B is expressed in units of β_1 then numerical value of P will be:

$$p_1 = \frac{A/\alpha_1}{B/\beta_1} \quad \text{in units of } \alpha_1 \text{ and } \beta_1$$

similarly if another quantity Q is given as $Q = A \times B$ then numerical value of Q is given as:

$$q_1 = \left(\frac{A}{\alpha_1} \times \frac{B}{\beta_1}\right) \text{ in units of } \alpha_1, \beta_1$$

The same idea can also be extended to products and quotients of powers of various physical quantities.

Example: In a certain system of units, acceleration due to gravity has a numerical value of 4.9 and velocity of light has a numerical value of 6×10^8. What are the units of length and time in multiples of meters and seconds in such a system?

Solution. Let unit of time be τ and that of length be λ. Now acceleration due to gravity is $9.8\ m\,s^{-2}$. The physical quantities remain unchanged on change of units thus:

$$g = 4.9\,(\lambda/\tau^2) = 9.8\,(m/s^2) \quad (1)$$

$$c = 6 \times 10^8\,(\lambda/\tau) = 3 \times 10^8\,(m/s) \quad (2)$$

From (1): $$2\,\frac{m}{s^2} = \frac{\lambda}{\tau^2} \quad (3)$$

From (2) $$\frac{1}{2}\,\frac{m}{s} = \frac{\lambda}{\tau} \quad (4)$$

YOUTUBE.COM/@THEPHYSICSSHELTER

④/③ gives $\tau = \frac{1}{4}$ sec, putting in ③, $\lambda = \frac{1}{8}$ m Ans.

→ One can readily verify the answer by converting back to standard units. $4.9\lambda/\tau^2 = \frac{4.9 \times (1/8\ m)}{(1/4\ sec)^2} = 9.8\ ms^{-2}$

Similarly $6\times10^8\ \lambda/\tau = 6\times10^8 \times (\frac{1}{8}\ m)/(\frac{1}{4}\ sec) = 3\times10^8\ ms^{-1}$.

EXAMPLE:

In a certain system of absolute units the acceleration produced by gravity in a body falling freely is denoted by 5, the kinetic energy of a 500 kg shot moving with velocity 400 metres per second is denoted by 2000 & its momentum by 100. ($g = 10\ ms^{-2}$)

The unit of length is :-

(A) 15 m (B) 50 m (C) 25 m (D) 100 m

The unit of time is :-

(A) 10 s (B) 20 s (C) 5 s (D) 15 s

The unit of mass is :-

(A) 200 kg (B) 400 kg (C) 800 kg (D) 1200 kg

YOUTUBE.COM/@THEPHYSICSSHELTER

Solution: Let us first evaluate KE and momentum in standard units.

$KE = \frac{1}{2} \times 500 \times 400^2 = 4\times10^7\ Kg\ m^2/s^2$

$P = 500 \times 400 = 2\times10^5\ kg\ m/s$

Now let the units of length, mass and time be respectively λ, μ and τ then according to the question.

$10\ ms^{-2} = 5\lambda\tau^{-2}$ ①

$4\times10^7\ Kg\ m^2 s^{-2} = 2000\ \mu\lambda^2\tau^{-2}$ ② and

$2\times10^5\ Kg\ ms^{-1} = 100\ \mu\lambda\tau^{-1}$ ③

Now ③²/② gives: $\frac{(2\times10^5)^2}{4\times10^7}\ Kg = \frac{10000}{2000}\ \mu$

$\Rightarrow \mu = 200\ Kg$ Ans.

Now ③/① gives: $20\ \mu\tau = 2\times10^4\ Kg\ sec$

$\Rightarrow \tau = \frac{10^3\ Kg\ sec}{\mu} = \frac{10^3\ kg\ ser}{200\ kg} = 5\ sec.$ Ans.

Now from ①, $\lambda = 2\ ms^{-2} \times \tau^2 = 2\ ms^{-2} \times (5sec)^2 = 50\ m$ Ans.

AMBARISH SIR
https://www.youtube.com/@ambarishphysics https://www.youtube.com/@thephysicsshelter

Thus $\lambda = 50\,m$, $\tau = 5\,sec$ and $\mu = 200\,kg$. The correct options are B, C and A respectively.

EXAMPLE:

In a new unit system, 1 unit of time is equal to 10 second, 1 unit of mass is 5 kg and 1 unit of length is 20 m. In this new system of units, 1 unit of energy is equal to

(A) 20 Joule

(B) $\frac{1}{20}$ Joule

(C) 4 Joule

(D) 16 Joule

Solution: Let μ, λ and τ be units of mass length and time in the new system. Now energy is $\frac{1}{2}mv^2$ i.e its units will be $\mu\lambda^2/\tau^2$. Thus our unit of energy becomes $5\,kg \times (20\,m)^2/(10\,sec)^2$ $= 20\,kg\,m^2/sec^2 = 20$ Joule. Thus correct option is A.

CLASSIFICATION OF UNITS: Units can be classified as fundamental, derived and supplementary. These are described as follows:

Fundamental or base units :

The units of fundamental quantities are called *base units*. In SI there are seven base units.

SI BASE QUANTITIES AND THEIR UNITS

S.No.	Physical quantity	SI unit	Symbol
1.	Length [L]	metre	m
2.	Mass [M]	kilogram	kg
3.	Time [T]	second	s
4.	Temperature [θ]	Kelvin	K
5.	Electric current [A]	ampere	A
6.	Luminous intensity [Cd]	candela	cd
7.	Amount of substance	mole	mol

→ Scientific community has agreed upon the quantities which will be considered as fundamental. As such there is nothing

YOUTUBE.COM/@THEPHYSICSSHELTER

"Fundamental" about them. For instance we could have chosen velocity instead of length as a fundamental quantity then the length would become a derived quantity having units of velocity x time. It is just a matter of convention that human civilization has agreed upon in the present times.

DERIVED QUANTITIES: The quantities that can be expressed in terms of basic or fundamental physical quantities are called derived quantities. For example momentum can be written as Mass x Velocity i.e Mass X length / Time (MLT^{-1}).

SUPPLEMENTARY QUANTITIES: These quantities are defined in addition to fundamental quantities. They are required for expressing angles and solid angles. Thus the two supplementary units are radians (rad) and steradians (sr). [Both are dimensionless].

DIMENSIONS: Derived quantities can be written in terms of products and quotients of powers of fundamental quantities. The powers of fundamental quantities in a derived unit is called the dimension of the fundamental quantity in the derived quantity. For example acceleration is expressed in units of length / time^2 i.e LT^{-2} thus dimensions of L and T in acceleration are 1 and -2. In general if unit of a physical quantity is expressed as $M^aL^bT^c$, a, b and c are called dimensions of mass, length and time in that physical quantity and $M^aL^bT^c$ is called dimensional formula for that physical quantity.

FINDING DIMENSIONAL FORMULA: If we know the dimensions of some quantities, we can express the dimensions of any other quantity which is related to these quantities through some physics formula. For instance suppose we want to find dimensions of ϵ_0 i.e permittivity of free space, we could use the formula :

YOUTUBE.COM/@THEPHYSICSSHELTER

$F = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2}$ thus ϵ_0 must have dimensions of $\frac{q_1 q_2}{r^2 F}$

Now charge has dimensions of (current x Time ie AT, and F has dimensions of MLT^{-2} (∵ mass x acceleration). Thus dimensions of ϵ_0:

$$[\epsilon_0] = \frac{[AT]^2}{L^2[MLT^{-2}]} = M^{-1}L^{-3}T^4A^2$$

DIMENSIONS OF IMPORTANT PHYSICAL QUANTITIES

Physical quantity	Dimensions	Physical quantity	Dimensions
Momentum	$M^1L^1T^{-1}$	Capacitance	$M^{-1}L^{-2}T^4A^2$
Calorie	$M^1L^2T^{-2}$	Modulus of rigidity	$M^1L^{-1}T^{-2}$
Latent heat capacity	$M^0L^2T^{-2}$	Magnetic permeability	$M^1L^1T^{-2}A^{-2}$
Self inductance	$M^1L^2T^{-2}A^{-2}$	Pressure	$M^1L^{-1}T^{-2}$
Coefficient of thermal conductivity	$M^1L^1T^{-3}K^{-1}$	Planck's constant	$M^1L^2T^{-1}$
Power	$M^1L^2T^{-3}$	Solar constant	$M^1L^0T^{-3}$
Impulse	$M^1L^1T^{-1}$	Magnetic flux	$M^1L^2T^{-2}A^{-1}$
Hole mobility in a semi conductor	$M^{-1}L^0T^2A^1$	Current density	$M^0L^{-2}T^0A^1$
Bulk modulus of elasticity	$M^1L^{-1}T^{-2}$	Young modulus	$M^1L^{-1}T^{-2}$
Potential energy	$M^1L^2T^{-2}$	Magnetic field intensity	$M^0L^{-1}T^0A^1$
Gravitational constant	$M^{-1}L^3T^{-2}$	Magnetic Induction	$M^1T^{-2}A^{-1}$
Light year	$M^0L^1T^0$	Electric Permittivity	$M^{-1}L^{-3}T^4A^2$
Thermal resistance	$M^{-1}L^{-2}T^3K$	Electric Field	$M^1L^1T^{-3}A^{-1}$
Coefficient of viscosity	$M^1L^{-1}T^{-1}$	Resistance	$ML^2T^{-3}A^{-2}$

YOUTUBE.COM/@THEPHYSICSSHELTER

EXAMPLE: Find dimensional formula for Magnetic field strength if fundamental quantities are assumed to be M, L, T and C (charge).

Solution: We will use the formula $F = ilB \Rightarrow B = F/il$

thus dimensions of B: $\frac{MLT^{-2}}{[CT^{-1}]L} = MT^{-1}C^{-1}$ Ans.

USING DIMENSIONAL FORMULA FOR UNIT CONVERSIONS: If a certain quantity has dimensions of $M^aL^bT^c$ and the units of mass, length and time are M_1, L_1 and T_1 and in another system

it is M_2 L_2 and T_2, further let numerical value in first system be N_1 and in second system, it is N_2 then:

$$N_1 M_1^a L_1^b T_1^c = N_2 M_2^a L_2^b T_2^c \Rightarrow N_2 = N_1 \left(\frac{M_1}{M_2}\right)^a \left(\frac{L_1}{L_2}\right)^b \left(\frac{T_1}{T_2}\right)^c.$$

→ This means value of dimensionless numbers will not change upon changing the system of units (such as Reynold's number, coefficient of friction or restitution or any other dimensionless ratio of physical quantities).

PRINCIPLE OF HOMOGENITY:

All the terms of an equation representing a physical process must have the same dimension. For example consider the following equation describing a spring mass system:

$$m\ddot{x} + k(x - x_0) = 0$$

Here each term has the same dimension as $m\ddot{x}$ i.e MLT^{-2}.

→ Arguments of trigonometric functions must be always dimensionless

→ Exponents and arguments of log must be dimensionless.

EXAMPLE: If $\frac{\alpha}{t^2} = Fv + \beta/x^2$ find dimensions of α and β if F is force, v is velocity, x is distance and t is time.

Solution: Dimensions of α must be same as that of Fvt^2 (∵ Homogeneity). Thus:

$$[\alpha] = [MLT^{-2}][LT^{-1}][T^2] = ML^2T^{-1} \text{ Ans.}$$

Now Dimensions of β must be same as that of Fvx^2 thus:

$$[\beta] = [MLT^{-2}][LT^{-1}][L^2] = ML^4T^{-3} \text{ Ans.}$$

EXAMPLE: Find dimensions of α and β if $\alpha = \frac{Fv^2}{\beta^2} \ln(2\pi\beta/v^2)$

Solution: Since argument of log is dimensionless, β must have dimension of v^2 i.e $[\beta] = [L^2T^{-2}]$

YOUTUBE.COM/@THEPHYSICSSHELTER

Now α has dimensions of $\frac{Fv^2}{\beta^2}$ i.e $\frac{Fv^2}{v^4}$ i.e Fv^{-2}

thus $[\alpha] = [MLT^{-2}][LT^{-1}]^{-2} = ML^{-1}T^{0}$ Ans.

Using Dimensional Analysis For Possible Correctness Of An Equation:

For an equation to be correct, all terms must have same dimensions (as described in previous section).

Example: Check whether the following equation could be possibly correct in some physical context:

$$P = 3Fv^2/(\pi^2 t^2 x)$$ (P: Pressure F: Force t: time, x: distance).

Solution: $[P] = [ML^{-1}T^{-2}]$

$$[RHS] = \frac{[MLT^{-2}][L^2T^{-2}]}{[T^2][L]} = ML^2T^{-6}$$

$[LHS] \neq [RHS]$ thus equation is dimensionally incorrect.

Using Dimensions To Derive Formulae: Rayleigh's Method:

If we know the physical quantities on which another physical quantity depends, we may be able to find the dependence proportionality (power) using dimensional analysis. This will be clear from following example:

Example: If we know that time period of a pendulum depends of mass of the bob, length of the string and acceleration due to gravity. Find the proportionality formula for the time period in terms of these quantities.

Solution: Let $T = m^a l^b g^c$ (m: mass, l: length of pendulum).

By dimensional homogeneity: $[T] = M^a L^b [LT^{-2}]^c$

equating dimensions of M, L and T: $a = 0$, $-2c = 1 \Rightarrow c = -1/2$

YOUTUBE.COM/@THEPHYSICSSHELTER

and $b + c = 0 \Rightarrow b = -c = +1/2$. This gives us the proportionality formula:

$$T \propto m^0 l^{1/2} g^{-1/2} \quad \text{i.e} \quad T \propto \sqrt{\frac{l}{g}}.$$

→ However dimensional analysis tells us nothing about proportionality constant.

→ If there were any fourth quantity involved expressible in terms of M, L and T, we would have another variable d but only three equations (corresponding to dimensions of M, L and T

→ This method fails to tell us if relationship involves multiple terms (it only tells powers of the quantities assuming dependence on a single term).

→ In general if an equation involves three dimensions a unique solution will exist only if physical variables involved are 4 (like in our case it was time period, mass, length and acc^n due to gravity) In general if k dimensions are involved, we will have unique solution if there are k+1 physical variables. (i.e. there will be a unique way of generating a dimensionless term using all k+1 quantities).

Buckingham Π Theorem: If n physical quantities are to be related to each other spanning a total of k dimensions then we can generate n−k dimensionless ratios out of them say $\Pi_1, \Pi_2, \Pi_3 \ldots \Pi_{n-k}$. Then the required physical relation must satisfy:

$$f(\Pi_1, \Pi_2, \Pi_3 \ldots, \Pi_{n-k}) = 0.$$

Our standard method for dimensional analysis is a special case of Buckingham Π theorem with $n = 4$ and $k = 3$. (Here only one dimensionless ratio is possible). If there are more than one dimensionless ratios possible, one may not be able to figure out exact proportionalities however one may still get valuable insights using the functional form generated using Buckingham Π theorem. We will illustrate use of Buckingham Π theorem using a very simple example at first.

Example: The displacement of a particle s moving with a constant acceleration a depends on initial velocity u and time (t)

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics https://www.youtube.com/@thephysicsshelter

Using Buckingham Π theorem, find the possible functional form relating the four variables.

Solution: The variables involved are $S:[L]$, $u:[LT^{-1}]$, $a:[LT^{-2}]$ and $t:[T]$. Dimensions involved are only two i.e L and T but variables are four. So we can make two basic dimensionless ratios. Let us choose the sets (S, t, u) and (S, t, a) to form dimensionless ratios.

Using first set: Let $S t^a u^b$ be the required dimensionless ratio then $L T^a (LT^{-1})^b = L^0 T^0 \Rightarrow 1+b=0 \Rightarrow b=-1$ and $a-b=0$ $\Rightarrow a=b=-1$

$\Rightarrow$ required ratio: $\Pi_1 = S/ut$.

Now from second set, $S t^a a^b = 0 \Rightarrow L T^a (LT^{-2})^b = L^0 T^0$

$\Rightarrow 1+b=0 \Rightarrow b=-1$ and $a-2b=0 \Rightarrow a=2b=-2$

$\Rightarrow$ required ratio is: $\Pi_2 = \frac{S}{at^2}$.

Thus required functional form is: $f(\Pi_1, \Pi_2) = 0$

$\Rightarrow f\left(\frac{S}{ut}, \frac{S}{at^2}\right) = 0$. Now this can be easily verified with the known result $s = ut + \frac{1}{2}at^2$

$\Rightarrow \frac{ut}{S} + \frac{1}{2}\frac{at^2}{S} - 1 = 0 = \frac{1}{\Pi_1} + \frac{1}{2\Pi_2} - 1 = 0$

→ While the Buckingham Π theorem does not give us exact equation, nonetheless, even knowing the functional form in terms of dimensionless ratios gives us a direction of doing the experiments so that exact functional form may be found by fitting a curve between Π_1 and Π_2 using experimental data.

→ Buckingham Π theorem is extensively used in fluid mechanics for deriving relation between various variables of fluid flow where analysis by exact laws of physics is not easy or possible, however finding the relationship is important for engineering applications. Often experiments are done on cheaper prototypes

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

before building the actual full scale functioning models).

EXPRESSING VARIOUS PHYSICAL QUANTITIES IN ALTERNATIVE DIMENSIONS:

As described earlier, we saw that seven fundamental physical quantities were arbitrarily chosen, however if we want we could call some other set as fundamental. For example instead of $[M]$, $[L]$ and $[T]$ we could choose Force $[F]$, velocity $[V]$ and time $[T]$ as fundamental. Then we could express dimensions of other physical quantities in terms of these fundamental dimensions. For a set of dimensions to be called fundamental, they should not be expressible in terms of each other. For instance $[F]$ can't be expressed in terms of $[V]$ and $[T]$ so we have a valid fundamental set. On the other hand we could not choose Energy $[E]$ momentum $[P]$ and velocity $[V]$ as fundamental set because $[E] = [P] \times [V]$.

EXAMPLE: If velocity $[V]$, force $[F]$ and time $[T]$ are chosen as fundamental quantities, express dimensions of (a) Mass (b) Energy in terms of chosen fundamental set.

Solution: (a) Let $[M] = [F]^a [V]^b [T]^c$ then

$[M] = [ML\bar{T}^2]^a [L\bar{T}^1]^b [T]^c$ thus, comparing exponents:

$a = 1$, $a + b = 0 \Rightarrow b = -1$ and $-2a - b + c = 0 \Rightarrow c = 2a + b = 1$

thus:

$$[M] = [F]^1 [V]^{-1} [T]^1$$ (∵ This can be seen as Impulse/Velocity).

(b) Let energy $[E] = [F]^a [V]^b [T]^c \Rightarrow [ML^2\bar{T}^2] = [ML\bar{T}^2]^a [L\bar{T}^1]^b [T]^c$

Now comparing coefficients:

$a = 1$, $a + b = 2 \Rightarrow b = 1$ and $-2a - b + c = -2 \Rightarrow c = -2 + 2a + b$

$\Rightarrow c = 1$ Thus required dimension is:

$$[E] = [FVT]$$ Ans. (This can also be seen as Power × Time).

YOUTUBE.COM/@THEPHYSICSSHELTER

Limitations Of Dimensional Analysis:

1. Dimensional analysis does not give any information about constants (multiplicative or additive) in the equations.
2. The standard Rayleigh's method (excluding Buckingham π), is useful only for multiplicative relations of the form $x^a y^b z^c$. For example it fails to give us a formula of the form $x+y+z$. A formula like $s = ut + \frac{1}{2}at^2$ can't be derived using dimensional analysis.
3. This method does not tell if there is dependence on other quantities in terms of Sine, Cos, Log etc type of functions.
4. Standard method (Rayleigh's method) works only if number of physical variables is one more than dimensions involved.

SI Units of Basic Quantities :

Base Quantity	SI Units		
	Name	Symbol	Definition
Length	metre	m	The metre is the length of the path traveled by light in vacuum during a time interval of 1/299, 792, 458 of a second (1983)
Mass	kilogram	kg	The kilogram is equal to the mass of the international prototype of the kilogram (a platinum-iridium alloy cylinder) kept at International Bureau of Weights and Measures, at Sevres, near Paris, France. (1889)
Time	second	s	The second is the duration of 9, 192, 631, 770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the cesium-133 atom (1967)
Electric Current	ampere	A	The ampere is that constant current which, if maintained in two straight parallel conductors of infinite length, of negligible circular cross-section, and placed 1 metre apart in vacuum, will produce between these conductors a force equal to 2×10^{-7} Newton per metre of length. (1948)
Thermodynamic Temperature	kelvin	K	The kelvin, is the fraction 1/273.16 of the thermodynamic temperature of the triple point of water. (1967)
Amount of Substance	mole	mol	The mole is the amount of substance of a system, which contains as many elementary entities as there are atoms in 0.012 kilogram of carbon-12. (1971)
Luminous Intensity	candela	cd	The candela is the luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency 540×10^{12} hertz and that has a radiant intensity in that direction of 1/683 watt per steradian (1979).

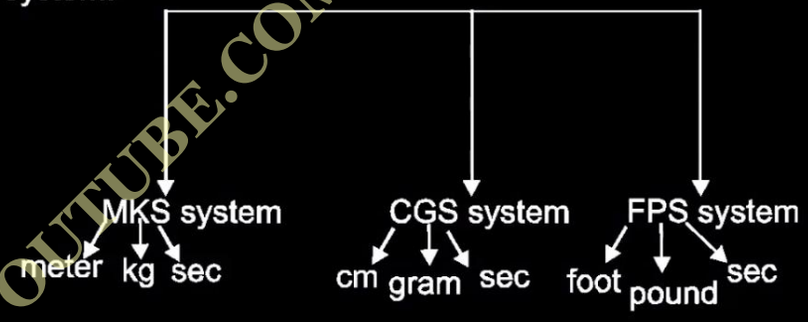
YOUTUBE.COM/@THEPHYSICSSHELTER

SI System: For the purpose of clear and good communication regarding measurements in the scientific community, and international organization CGPM (General conference on weights and measures) decided the standard units which are internationally accepted. These units are called SI units (International system of units). The accepted definitions of various standard units according to this system are given in the table above.

→ In 2019 four out of 7 base units in SI system were redefined in terms of natural constants rather than human artifacts such as standard kilogram. These definitions are not yet updated in the NCERT. For instance, second was defined in terms of transition between two hyperfine levels of caesium atom, Accordingly, metre was defined in terms of velocity of light and oscillation frequency of caesium atom, Kilogram was defined in terms of planck's constant, velocity of light and caesium oscillation frequency and Ampere was defined in terms of electronic charge and caesium oscillation frequency.

Other Systems of Units :

If a quantity involves only length, mass and time (quantities in mechanics), then its unit can be written in MKS, CGS or FPS system.



Prior to SI system, some popular systems of units were MKS, CGS and FPS as described in the chart above. MKS is a subset of SI system. FPS system is still used in Britain and US, however working in these systems is crazy due to weird conversion factors. Gradually the textbooks from these countries are also adopting SI (or publishing SI versions of textbooks).

YOUTUBE.COM/@THEPHYSICSSHELTER

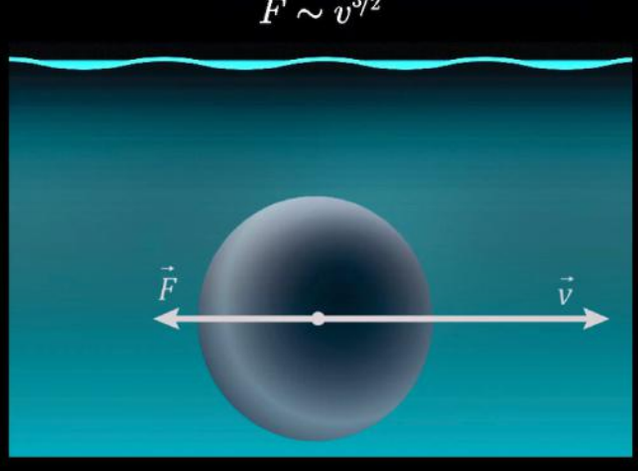
Prefixes Used In SI System :

The Metric System Prefixes				
Prefix	Symbol	Decimal Value	Scientific	Colloquial
quecto	q	0.000 000 000 000 000 000 000 000 000 001	10^{-30}	nonillionth
ronto	r	0.000 000 000 000 000 000 000 000 001	10^{-27}	octillionth
yocto	y	0.000 000 000 000 000 000 000 001	10^{-24}	septillionth
zepto	z	0.000 000 000 000 000 000 001	10^{-21}	sextillionth
atto	a	0.000 000 000 000 000 001	10^{-18}	quintillionth
femto	f	0.000 000 000 000 001	10^{-15}	quadrillionth
pico	p	0.000 000 000 001	10^{-12}	trillionth
nano	n	0.000 000 001	10^{-9}	billionth
micro	μ	0.000 001	10^{-6}	millionth
milli	m	0.001	10^{-3}	thousandth
centi	c	0.01	10^{-2}	hundredth
deci	d	0.1	10^{-1}	tenth
-	-	1	10^{0}	one
deka	da	10	10^{1}	ten
hecto	h	100	10^{2}	hundred
kilo	k	1 000	10^{3}	thousand
mega	M	1 000 000	10^{6}	million
giga	G	1 000 000 000	10^{9}	billion
tera	T	1 000 000 000 000	10^{12}	trillion
peta	P	1 000 000 000 000 000	10^{15}	quadrillion
exa	E	1 000 000 000 000 000 000	10^{18}	quintillion
zetta	Z	1 000 000 000 000 000 000 000	10^{21}	sextillion
yotta	Y	1 000 000 000 000 000 000 000 000	10^{24}	septillion
ronna	R	1 000 000 000 000 000 000 000 000 000	10^{27}	octillion
quetta	Q	1 000 000 000 000 000 000 000 000 000 000	10^{30}	nonillion

The various prefixes used before SI units are listed out in the table above. Out of this, student is advised to remember prefixes in the dotted rectangle.

YOUTUBE.COM/@THEPHYSICSSHELTER

Example : In a certain fluid measurements indicate that drag force is proportional to $v^{3/2}$ where v is instantaneous velocity. Two spheres with radii r_1 and r_2 are pushed with equal velocity in the fluid such that $r_2 = 2r_1$. Determine the ratio of drag forces $\gamma = F_2/F_1$ where F_2 is drag force on second sphere and F_1 on the first sphere. Make suitable assumptions.



Solution: Let us assume that drag force depends on velocity, viscosity coefficient, density of liquid and radius of the ball. Thus:

$$[F] = [\eta]^x [r]^y [\rho]^z [v]^{3/2}$$

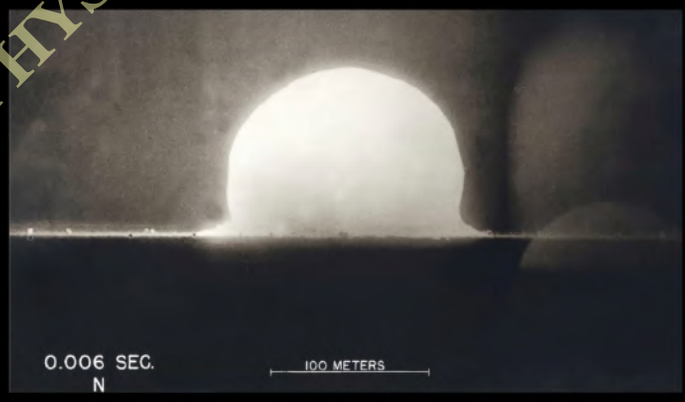
$$\Rightarrow M L T^{-2} = [M L^{-1} T^{-1}]^x [L]^y [M L^{-3}]^z [L T^{-1}]^{3/2}$$

Thus $x + z = 1$, $-x + y - 3z + \frac{3}{2} = 1$ and $-x - \frac{3}{2} = -2$

Solving we get $F \propto \eta^{1/2} r^{3/2} \rho^{1/2} v^{3/2}$

Thus $\gamma = \frac{F_2}{F_1} = \left(\frac{r_2}{r_1}\right)^{3/2} = 2^{3/2} \approx 2.82$ times.

EXAMPLE: A picture of the first nuclear bomb explosion shows that at a moment $t = 0.006$ sec after the explosion, the radius of the explosion cloud was 80 metres. A spy satellite captures a picture of a similar nuclear test by North Korea and it is found that after $t = 0.012$ sec from the explosion, the cloud radius is 160 metres. How many times more energy is released in this explosion assuming density of air is $1.2\ kg/m^3$. Make suitable assumptions.



Solution: Assuming that energy released depends on radius of cloud, time and density of air:

$$E \propto r^x t^y \rho^z$$

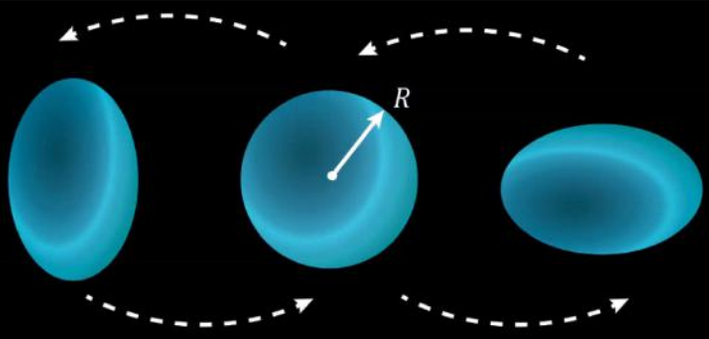
$$\Rightarrow [M L^2 T^{-2}] = L^x T^y [M L^{-3}]^z \Rightarrow z = 1,\ x - 3z = 2 \Rightarrow x = 5 \text{ and } y = -2.$$

Thus $E \propto r^5 t^{-2} \rho^1$ thus $\frac{E_2}{E_1} = \left(\frac{r_2}{r_1}\right)^5 \times \left(\frac{t_1}{t_2}\right)^2$

(Assuming same air density), thus: $E_2 = E_1 \times 2^5 \times \frac{1}{2^2} = 8E_1$ Ans.

YOUTUBE.COM/@THEPHYSICSSHELTER

EXAMPLE: A small water droplet floating in a space station usually has a spherical form which changes to ellipsoidal with some regular time period T. Find the ratio of time period of oscillations of drops with radii ratio $r_2/r_1 = 2$ (find T_2/T_1). Make suitable assumptions.



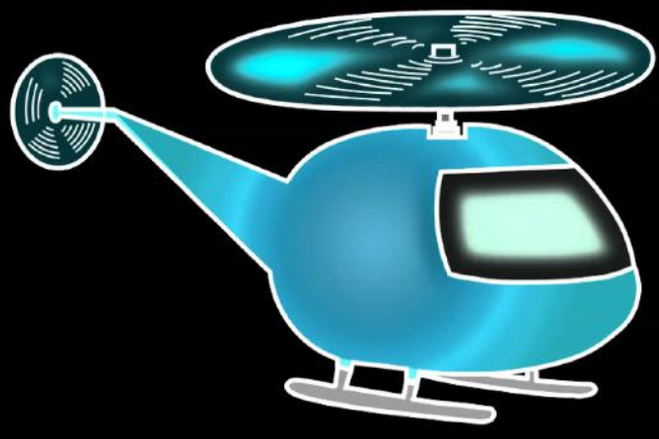
Solution: Let us assume time period to depend on radius, surface tension and density. Thus:

$$T \propto r^x \gamma^y \rho^z \Rightarrow [T] = L^x [MT^{-2}]^y [ML^{-3}]^z$$

$\Rightarrow -2y = 1 \Rightarrow y = -1/2$, $y + z = 0 \Rightarrow z = -y = 1/2$ and $x - 3z = 0 \Rightarrow x = 3/2$ thus:

$$T \propto r^{3/2} \gamma^{-1/2} \rho^{1/2} \Rightarrow \frac{T_2}{T_1} = 2^{3/2} = 2\sqrt{2} \approx 2.83 \text{ Ans.}$$

EXAMPLE: A certain helicopter is able to hover if power output of its engine is P_1. If an exact replica is made with all linear dimensions doubled. What power P_2 will be required for hovering. Make suitable assumptions.



Solution: We expect the Power to depend on mass of the helicopter (which in turn depends on average density and linear dimension), propeller radius, density of air and average density of helicopter. (Too many unknowns). We know that density of helicopter and that of air will have some constant ratio so we can just work with air density. Similarly mass just depends on linear dimension and density, so we will not consider mass as the variable (we will anyway consider linear dimension and density). In addition, we expect a dependence on acceleration due to gravity.

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics
https://www.youtube.com/@thephysicsshelter

Thus let $P \propto l^x \rho^y g^z \Rightarrow [ML^2T^{-3}] = [L]^x [ML^{-3}]^y [LT^{-2}]^z$

$\Rightarrow y = 1,\ x - 3y + z = 2,\ -2z = -3 \Rightarrow z = 3/2,\ x = 7/2$

Thus:

$P \propto l^{7/2} \rho\, g^{3/2}$. Thus doubling the linear dimensions will increase the required power to $2^{7/2} \approx 11.3$ times Ans.

EXAMPLE:

Two identical amounts of explosives are burst underwater at locations A and B where hydrostatic pressures are P_A and P_B respectively ($P_B = 1.3 P_A$). A hollow space occurring due to underwater explosion expands and shrinks with some regular time period T, which does not depend on size of the hollow bubble. Assume that sizes of the hollows are much smaller than depths of explosions A and B determine the ratio of oscillation time periods T_B / T_A. Make suitable assumptions.



YOUTUBE.COM/@THEPHYSICSSHELTER

Solution: Let us assume that time period depends on, Pressure, fluid density and energy of the explosion. (Why did we choose these parameters? Because dimensions involved are 3 and therefore according to Rayleigh's method, exact solution will not exist for more than 4 physical quantities). Thus:

$$T \propto P^x E^y \rho^z$$

i.e $[T] = [ML^{-1}T^{-2}]^x [ML^2T^{-2}]^y [ML^{-3}]^z \Rightarrow -2x - 2y = 1,\ x + y + z = 0$

and $-x + 2y - 3z = 0 \Rightarrow x = -5/6,\ y = 1/3,\ z = 1/2$ thus:

$$T \propto \frac{E^{1/3} \rho^{1/2}}{P^{5/6}}. \text{ Thus, } \frac{T_B}{T_A} = \left(\frac{P_A}{P_B}\right)^{5/6} = \left(\frac{1}{1.3}\right)^{5/6} = 0.8 \text{ Ans.}$$

AMBARISH SIR

https://www.youtube.com/@ambarishphysics https://www.youtube.com/@thephysicsshelter